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FSC 5365

PREPARING ACTIVITY:
NAVY - OS
Other Govt
ARMY - AR
AIR FORCE - 99

PROCUREMENT SPECIFICATION
MIL-R-21248

INTERNATIONAL INTEREST

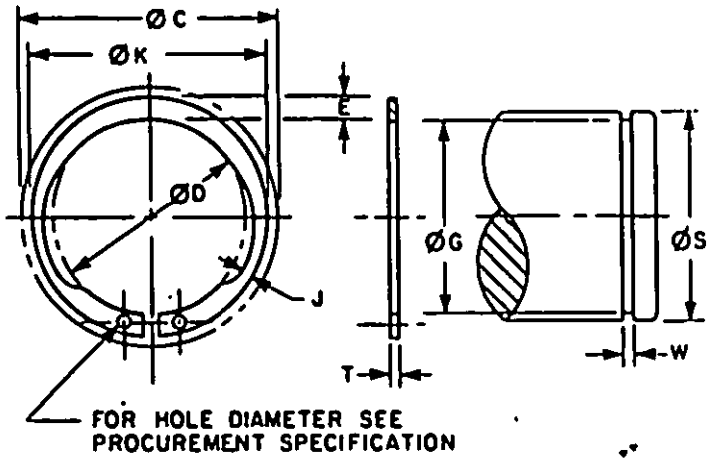
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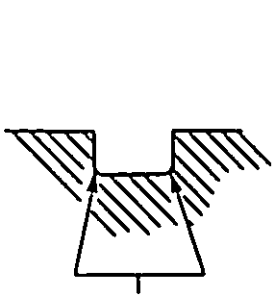
TITLE
RING, RETAINING, EXTERNAL, INVERTED (TAPERED SECTION TYPE)

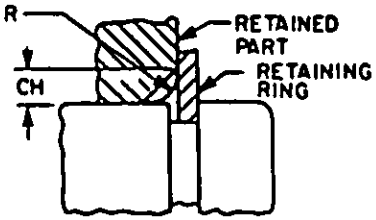
MILITARY STANDARD
MS16626

PAGE 1 OF 5 PAGES

(B) ENTIRE STANDARD REVISED







GROOVE BOTTOM RADII (MAX)
SHAFT DIAS.: .500 TO 1.000 • .005
1.062 TO 2.000 • .010
2.062 AND OVER • .015

ENLARGED DETAILS

TABLE I. DIMENSIONS

ØB SHAFT (REF)		ØD FREE		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		ØK 2/	ØC 3/	R 2/ CH 2/ OF RETAINED PART (REF)	
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX
.500	12.7	.461		.080		.041		.035		.460	+.002 -.001 FIM 4/	.040		.640	.670	.051	.032
.562	14.3	.521		.088	±.004	.043	±.004	.035		.530		.040		.715	.750	.057	.036
.594	15.1	.550		.092		.046		.035		.559		.040		.750	.790	.059	.037
.625	15.9	.579		.096		.048		.035		.588		.040		.790	.830	.062	.039
.672	17.1	.621		.102		.051		.035		.631		.040		.850	.890	.065	.041
.688	17.5	.635		.104		.052		.042		.646		.047		.870	.910	.066	.042
.750	19.0	.693	+.005	.112		.056		.042		.704		.047	+.003	.945	.990	.071	.045
.781	19.8	.722	-.010	.116		.057		.042		.733		.047	-.000	.980	1.040	.073	.046
.812	20.6	.751		.120	±.005	.060	±.005	.042	±.002	.762	±.003 -.001 FIM 4/	.047		1.020	1.080	.076	.048
.875	22.2	.810		.128		.064		.042		.821		.047		1.095	1.150	.080	.051
.938	23.8	.867		.136		.068		.042		.882		.047		1.170	1.230	.086	.054
.984	25.0	.925		.144		.072		.042		.926		.047		1.240	1.300	.091	.057
1.000	25.4	.925		.144		.072		.042		.940		.047		1.250	1.310	.091	.057
1.062	27.0	.982		.147		.073		.050		.998		.056		1.310	1.380	.092	.058
1.125	28.6	1.041		.150		.075		.050		1.059		.056		1.380	1.450	.093	.059
1.188	30.2	1.098		.153		.076		.050		1.118		.056		1.450	1.520	.094	.059
1.250	31.7	1.156		.157		.079		.050		1.176		.056	+.004	1.520	1.590	.096	.060
1.312	33.3	1.214	+.010	.161		.080		.050		1.232		.056	-.000	1.580	1.660	.097	.061
1.375	34.9	1.272	-.015	.165	±.006	.082	±.006	.050		1.291	±.004 -.003 FIM 4/	.056		1.650	1.730	.098	.061
1.438	36.5	1.333		.169		.085		.050		1.350		.056		1.715	1.800	.100	.063
1.500	38.1	1.387		.173		.086		.050		1.406		.056		1.775	1.870	.100	.063

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DD FORM 672-1 (REVISED)

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TABLE I. DIMENSIONS, -CONTINUED

ØS SHAFT (REF)		ØD FREE		E LARGE SECTION HEIGHT		J SMALL SECTION HEIGHT		T 1/ THICKNESS		ØC RECOMMENDED GROOVE (REF)		M WIDTH		ØK 2/	ØC 3/	R 5/ OR CH 5/ OF RETAINED PART (REF)	
INCH	MM	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX		MAX	MAX
1.562	39.7	1.446		.178	±.006	.089	±.006	.062		1.468		.068		1.850	1.950	.106	.066
1.750	44.4	1.637	+.013	.196		.098		.062		1.650		.068	+.004	2.070	2.180	.116	.073
1.772	45.0	1.637	-.020	.196		.098		.062		1.669		.068	-.000	2.090	2.200	.116	.073
1.812	46.0	1.675		.199		.100		.062		1.708	±.005 .005 FIM 4/	.068		2.130	2.240	.117	.074
1.969	50.0	1.819		.212		.106		.062		1.857		.068		2.310	2.430	.124	.078
2.000	50.8	1.850		.216		.108		.062		1.886		.068		2.350	2.470	.127	.080
2.125	54.0	1.993		.229		.117		.078		2.003		.086		2.490	2.620	.133	.084
2.156	54.8	1.993	+.015	.229		.117		.078	±.003	2.032		.086		2.520	2.650	.133	.084
2.500	63.5	2.313	-.025	.259	±.007	.130	±.007	.078		2.360		.086		2.910	3.050	.151	.095
2.875	73.0	2.659		.290		.145		.093		2.721		.103		3.330	3.490	.170	.107
3.156	80.1	2.920	+.020	.316		.159		.093		2.986	±.006	.103	+.005	3.650	3.820	.185	.116
3.500	88.8	3.237	-.030	.345	±.008	.173	±.008	.109		3.316	.006 FIM 4/	.120	-.000	4.030	4.220	.202	.127
3.938	100.0	3.642		.368		.183		.109		3.734		.120		4.500	4.710	.212	.133

1/ T = THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e., ±.002 SHOULD BE +.004/-.002.
2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.
3/ K - MAXIMUM DIAMETER WHEN THE RING IS PROPERLY SEATED IN THE GROOVE. (DESIGN REFERENCE DIMENSION).
4/ C - ACTUAL CLEARANCE DIAMETER WHEN THE RING IS SPRUNG OVER THE SHAFT, PRIOR TO INSTALLATION INTO THE GROOVE. (DESIGN REFERENCE DIMENSION).
5/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS FOR RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS ARE TABULATED ON PAGE 6.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 4 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:
(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568 OR ASTM A682.
(b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700).
(c) BERYLLIUM COPPER ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B 194.

3. HARDNESS:

TABLE II. HARDNESS

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.500 TO .672	50-55 HRC	63-69.5 HR30N	56.5-62 HR30N
.688 TO .812	48-53 HRC	63-69.5 HR30N	56.5-62 HR30N
.875 TO 1.500	48-53 HRC	44-51 HRC	37-43 HRC
1.562 TO 3.437	46-51 HRC	44-51 HRC	37-43 HRC
3.500 & OVER	44-49 HRC	44-51 HRC	37-43 HRC

4. PROTECTIVE FINISH OR SURFACE TREATMENT:
a. CARBON STEEL - SHALL BE AS SPECIFIED (SEE TABLE III OR IV):
1. CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5.
2. ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn5, OR ASTM B695, TYPE II, CLASS 5.
3. PHOSPHATE COAT IN ACCORDANCE WITH DOD-P-16232, TYPE Z, CLASS 2.

b. CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH QQ-P-35.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV.

EXAMPLE: MS16626-1100 IS THE PART NUMBER FOR A CARBON STEEL CADMIUM PLATE, EXTERNAL, INVERTED RETAINING RING FOR USE ON A 1.000 SHAFT DIAMETER.

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

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P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY-OS Other Cust ARMY-AR AIR FORCE-99		RING, RETAINING, EXTERNAL, INVERTED (TAPERED SECTION TYPE)	MS16626
PROCUREMENT SPECIFICATION MIL-R-21248	SUPERSEDING		PAGE 3 OF 6

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PROCUREMENT SPECIFICATION
MIL-R-21248

INTERNATIONAL INTEREST

SUPERSEDES:

TITLE
RING, RETAINING, EXTERNAL, INVERTED (TAPERED SECTION TYPE)

MILITARY STANDARD

MS16626

4 6

TABLE III. DASH NUMBERS FOR MS16626

Øs SHAFT (REF)	CARBON STEEL, CADMIUM PLATE 1/	CARBON STEEL, ZINC COAT 1/	CARBON STEEL, PHOSPHATE COAT 1/	STEEL, CORROSION RESISTANT	BERYLLIUM COPPER 1/
	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.500	-1050	-2050	-3050	-4050	-5050
.562	-1056	-2056	-3056	-4056	-5056
.594	-1059	-2059	-3059	-4059	-5059
.625	-1062	-2062	-3062	-4062	-5062
.672	-1067	-2067	-3067	-4067	-5067
.688	-1068	-2068	-3068	-4068	-5068
.750	-1075	-2075	-3075	-4075	-5075
.781	-1078	-2078	-3078	-4078	-5078
.812	-1081	-2081	-3081	-4081	-5081
.875	-1087	-2087	-3087	-4087	-5087
.935	-1093	-2093	-3093	-4093	-5093
.984	-1098	-2098	-3098	-4098	-5098
1.000	-1100	-2100	-3100	-4100	-5100
1.062	-1106	-2106	-3106	-4106	-5106
1.125	-1112	-2112	-3112	-4112	-5112
1.188	-1118	-2118	-3118	-4118	-5118
1.250	-1125	-2125	-3125	-4125	-5125
1.312	-1131	-2131	-3131	-4131	-5131
1.375	-1137	-2137	-3137	-4137	-5137
1.438	-1143	-2143	-3143	-4143	-5143
1.500	-1150	-2150	-3150	-4150	-5150
1.562	-1156	-2156	-3156	-4156	-5156
1.750	-1175	-2175	-3175	-4175	-5175
1.772	-1177	-2177	-3177	-4177	-5177
1.812	-1181	-2181	-3181	-4181	-5181
1.969	-1196	-2196	-3196	-4196	-5196
2.000	-1200	-2200	-3200	-4200	-5200
2.125	-1212	-2212	-3212	-4212	
2.156	-1215	-2215	-3215	-4215	
2.500	-1250	-2250	-3250	-4250	
2.875	-1287	-2287	-3287	-4287	
3.156	-1315	-2315	-3315	-4315	
3.500	-1350	-2350	-3350	-4350	
3.938	-1393	-2393	-3393	-4393	

TABLE IV. SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS)

Øs SHAFT (REF)	INACTIVE	SUBSTITUTE	SUBSTITUTE	SUBSTITUTE
	CARBON STEEL	CARBON STEEL, CADMIUM PLATE 1/	CARBON STEEL, ZINC COAT 1/	CARBON STEEL, PHOSPHATE COAT 1/
	MS16626	MS16626	MS16626	MS16626
.500	-50	-1050	-2050	-3050
.562	-56	-1056	-2056	-3056
.594	-59	-1059	-2059	-3059
.625	-62	-1062	-2062	-3062
.672	-67	-1067	-2067	-3067
.688	-68	-1068	-2068	-3068
.750	-75	-1075	-2075	-3075
.781	-78	-1078	-2078	-3078
.812	-81	-1081	-2081	-3081
.875	-87	-1087	-2087	-3087
.935	-93	-1093	-2093	-3093
.984	-98	-1098	-2098	-3098
1.000	-100	-1100	-2100	-3100
1.062	-106	-1106	-2106	-3106
1.125	-112	-1112	-2112	-3112
1.188	-118	-1118	-2118	-3118
1.250	-125	-1125	-2125	-3125
1.312	-131	-1131	-2131	-3131
1.375	-137	-1137	-2137	-3137
1.438	-143	-1143	-2143	-3143
1.500	-150	-1150	-2150	-3150
1.562	-156	-1156	-2156	-3156
1.750	-175	-1175	-2175	-3175
1.772	-177	-1177	-2177	-3177
1.812	-181	-1181	-2181	-3181
1.969	-196	-1196	-2196	-3196
2.000	-200	-1200	-2200	-3200
2.125	-212	-1212	-2212	-3212
2.156	-215	-1215	-2215	-3215
2.500	-250	-1250	-2250	-3250
2.875	-287	-1287	-2287	-3287
3.156	-315	-1315	-2315	-3315
3.500	-350	-1350	-2350	-3350
3.938	-393	-1393	-2393	-3393

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (232°C).

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RECOMMENDED DESIGN LIMITATIONS AND USAGE:

(a) INTENDED USE - TO PROVIDE UNIFORM PROTRUDING SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS. TAPERED DESIGN PRINCIPLE PERMITS RINGS TO MAINTAIN PRACTICALLY CONSTANT CIRCULARITY, AND FIT SECURELY AGAINST BOTTOM OF GROOVE, COUNTERACTING CONSIDERABLE CENTRIFUGAL FORCE (SEE CHART BELOW). ESPECIALLY SUITED FOR LOCATING AND RETAINING MACHINE PARTS HAVING CURVED ABUTTING SURFACES. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION.
LIMITATION ON USE - THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED, (PROVIDING GROOVE HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

TABLE V. APPROXIMATE SAFETY RPM LIMITS

Ø SHAFT (INCHES)		.500	1	2	3	4
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	40,000	20,000	10,000	6,700	5,000
BERYLLIUM COPPER	RPM LIMIT	25,000	13,000	6,400		

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RING. ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$$P = \frac{2}{3}\,\frac{\pi S T X}{F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFTEY FACTOR, F=4, IS RECOMMENDED, SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES, BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL -

$$P = \frac{1}{2}\,\frac{\pi S d Y}{F}$$

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F=2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT -

$$Z = 3d$$

WHERE:
Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

(e) DIFFERENTIAL ROTATION -
THE CONDITIONS UNDER WHICH A RETAINING RING MAY BE USED WHEN ADJACENT PARTS ROTATE RELATIVE TO IT, FALL INTO TWO CATEGORIES -

(1) WHERE NO THRUST IS EXERTED BY ADJACENT PART: IN THIS CASE, DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RING BECAUSE NO FRICTIONAL TORQUE IS EXERTED BY THE MACHINE PART ON THE RING.

(2) CONSIDERATION MUST BE GIVEN TO THE MAGNITUDE OF THE THRUST INVOLVED. THE FRICTION MOMENT MAY NOT EXCEED THE BENDING MOMENT WHICH THE RING CAN TOLERATE WITHOUT RELEASING ITS PRESSURE AGAINST THE BOTTOM OF THE GROOVE, FORMULATED AS FOLLOWS:

$$fPN \leq \frac{sTE^2}{16} \quad \text{OR} \quad P \leq \frac{sTE^2}{16fN}$$

WHERE:
P = ALLOWABLE THRUST LOAD EXERTED BY ADJACENT PART (POUNDS)
f = COEFFICIENT OF FRICTION
s = WORKING STRESS OF RING UNDER MAXIMUM EXPANSION (PSI) 2/
T = RING THICKNESS (INCHES)
E = GREATEST WIDTH SECTION OF RING (INCHES)
N = NEUTRAL RING DIAMETER (INCHES) = FREE DIAMETER PLUS 3/4 E DIMENSION

IN SUCH CASES WHERE DIFFERENTIAL ROTATION OCCURS, THE CALCULATION SHOULD BE BASED ON THE MAXIMUM POSSIBLE VALUE OF THE COEFFICIENT OF FRICTION.

1/ X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER.
2/ s = 250,000 PSI WORKING STRESS FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
s = 160,000 PSI WORKING STRESS FOR RINGS OF BERYLLIUM COPPER.

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P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY-OS Other Cust ARMY-AR AIR FORCE-99		RING, RETAINING, EXTERNAL, INVERTED (TAPERED SECTION TYPE)	MS16626
PROCUREMENT SPECIFICATION MIL-R-21248	SUPERSEDES:		PAGE 5 OF 6

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(f) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ - FOR THE RING (INCH POUNDS). ABUTTING COMPONENTS TO HAVE SHARP CORNERS

$I_G = \frac{Pd}{2}$ - FOR THE GROOVE (INCH POUNDS)

WHERE:
- P = ALLOWABLE THRUST LOAD OF RINGS OR GROOVES (POUNDS)
- T = RING THICKNESS (INCHES)
- I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
- d = GROOVE DEPTH (INCHES)
- I_R = IMPACT CAPACITY OF RING (INCH POUNDS)

(g) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED =
WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON PAGE 5 WILL APPLY. THE CORNER RADII AND CHAMFERS LISTED ON PAGES 1 AND 2 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE VI.
IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY, IN ACCORDANCE WITH THE FOLLOWING FORMULAS.

$P^1 = \frac{P\,CH}{CH^1}$ OR $P^1 = \frac{PR}{R^1}$

WHERE:
- P^1 = NEW ALLOWABLE THRUST LOAD
- P = LISTED ALLOWABLE THRUST LOAD
- CH^1 = NEW (SMALLER) CHAMFER
- CH = LISTED CHAMFER
- R^1 = NEW (SMALLER) CORNER RADIUS
- R = LISTED CORNER RADIUS

LIMIT LOADS LISTED BELOW ARE BASED ON RINGS OF STEEL (WORKING STRESS 250,000 PSI) AND OF BERYLLIUM COPPER (WORKING STRESS 180,000 PSI). IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA ON PAGE 5 ARE LESS, THEN THEY SHOULD BE USED.

TABLE VI. LIMIT LOADS

NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS.	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.500	.672	680 LB	500 LB
.688	1.000	1000 LB	700 LB
1.062	1.500	1460 LB	1050 LB
1.563	2.000	2250 LB	1600 LB
2.125	2.500	3750 LB	
2.875	3.156	5500 LB	
3.500	3.938	7850 LB	

USER ACTIVITIES:
ARMY - AT, ER, ME
NAVY - AS, MC, SH, YD

REVIEWER ACTIVITIES:
ARMY - AV, GL, MI
AIR FORCE - 82
DLA - IS

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APPROVED 11 DEC 1958 REVISED (8) FOR CHANGES SEE PAGES 1 THRU 6

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